

ProLoop: Progress-Aware Closed-Loop Framework via Structured State Relations for Long-Horizon Robotic Task Planning

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Abstract—Long-horizon robotic tasks are commonly decomposed into semantic subgoals to simplify planning. However, subgoal progress in LLM-based planning is typically inferred implicitly from interaction histories. Inaccurate estimates can misguide execution and allow local errors to propagate. To address this, we propose ProLoop, a Progress-Aware Closed-Loop framework that models subgoal progress from structured relations between desired and execution states. A learned State-Relation Monitor assesses these relations, which are then aggregated to derive subgoal progress as ongoing, deviated, or completed. The resulting progress guides execution adaptation, enabling reliable progress-aware control throughout execution. Experiments on AgentBoard ALFWorld and a real-world robotic platform show that ProLoop improves success rate by 12.94% overall and by 15.46% on Hard tasks, while reducing environment interactions by 32.3%. Project page(anonymized): <https://proloopplanning.github.io/>

I. INTRODUCTION

Recent advances in robot task planning increasingly leverage large language models (LLMs) to generate coherent multi-step plans [1], [2]. Despite strong performance on complex tasks, LLM-based planning remains susceptible to hallucinations and logical inconsistencies [3], [4]. To manage long-horizon planning complexity, hierarchical methods decompose tasks into semantic subgoals, allowing each stage to focus on a simpler intermediate objective [5]–[7]. However, during subgoal execution, progress is still commonly inferred implicitly by the LLM from accumulated interaction trajectories and multi-source information [8]. Inaccurate progress estimates can therefore misguide execution, allowing local errors to accumulate and propagate across subsequent subgoals.

Building on these observations, recent studies [8]–[10] address this problem by introducing explicit checks at selected checkpoints to assess the current execution state and then support replanning (see Fig. 1). These checks form a feedback loop that reduces reliance on implicit progress inference by the LLM. However, such checkpoint-based feedback cannot timely respond to changes in subgoal progress, since reassessment occurs only at the next checkpoint. During robotic execution, this delay can allow erroneous actions to alter the environment state, making recovery costly or even infeasible [11], [12]. We therefore ask: **How can robots maintain reliable progress awareness throughout subgoal execution to enable timely closed-loop adaptation?**

A useful intuition comes from how people perform everyday tasks. People assess progress by relating the current state to the intended outcome before deciding what to do next. Following this intuition, we define robotic subgoal progress by aggregating structured relations between desired subgoal

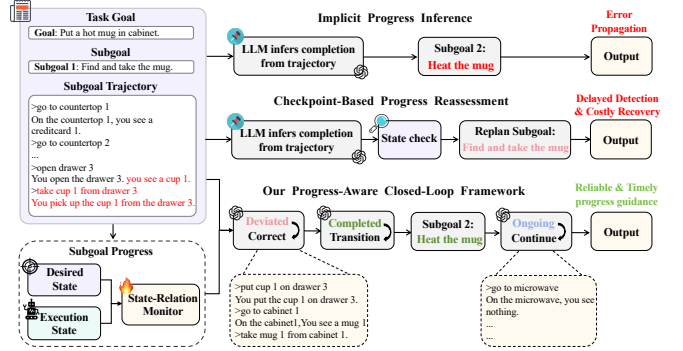


Fig. 1. Compared with existing approaches, ProLoop explicitly derives subgoal progress from the relation between desired and execution states, distinguishes *ongoing*, *deviated*, and *completed* execution, and uses the resulting progress to support timely closed-loop adaptation.

conditions and the evolving execution state into *ongoing*, *deviated*, or *completed*.

This formulation raises two key challenges. **Constructing comparable desired and execution states.** A semantic subgoal specifies its desired outcome in natural language, whereas the execution state must be reconstructed from evolving actions, observations, perception, and contextual information. Making the two states comparable is difficult, but essential for accurate subgoal progress assessment. **Using progress to guide execution without sacrificing LLM flexibility.** Even with reliable progress assessment, different progress states call for different responses. Excessive intervention during ongoing execution may disrupt useful LLM reasoning and action generation, whereas insufficient intervention after deviation may allow errors to persist. Moreover, completion should be leveraged beyond stage termination, as its associated outcomes must be selectively transferred to support subsequent planning.

To address these challenges, we propose ProLoop, a Progress-Aware Closed-Loop framework for long-horizon robotic task planning. For each active subgoal generated by the planner, **Structured State Construction** first converts the subgoal’s desired outcome and evolving execution information into comparable desired and execution states under a shared task-state schema. A learned **State-Relation Monitor** then assesses these condition-level relations and derives the active subgoal progress as ongoing, deviated, or completed. The resulting progress drives **Progress-Guided Adaptation**: ongoing execution preserves the underlying LLM’s reasoning process, deviation triggers targeted correction, and completion permits transition to the next subgoal. When progress

indicates completion, **Completion-Gated Memory Update** uses planner-generated retention cues to selectively summarize task-relevant execution information for subsequent planning.

To assess the effectiveness of ProLoop, we conduct experiments on the AgentBoard [13] ALFWorld household planning benchmark, which contains six task categories with easy and hard difficulty levels. Empirical results demonstrate improved overall performance, with especially substantial gains on hard tasks. We further validate ProLoop on a physical robot. Our contributions are threefold:

- 1) We propose **ProLoop**, a unified framework integrating structured state construction, state-relation monitoring, progress-guided adaptation, and completion-gated memory update for long-horizon tasks.
- 2) We define explicit subgoal progress through structured relations between desired and execution states, distinguishing *ongoing*, *deviated*, and *completed* progress.
- 3) We demonstrate that ProLoop improves task success rate and progress rate on ALFWorld while requiring fewer environment interactions. We further validate its feasibility on a physical robot.

II. RELATED WORK

A. LLMs for Task Planning

Recent studies have leveraged the reasoning and long-horizon decision-making capabilities of LLMs for task planning [3]. LID [14] formulates observations, goals, and interaction histories as sequential representations for policy prediction, while ReAct [15] interleaves reasoning and action generation. Code-based methods, including Code as Policies [16] and ProgPrompt [1], further represent plans as executable programs to improve structural consistency and interpretability. In embodied settings, SayCan [17] grounds language-based reasoning in robot affordances, and LLM-Planner [18] extends LLM-based planning to embodied decision-making tasks. These methods generate and ground multi-step plans, but long-horizon dependencies and growing interaction histories motivate more structured task organization across successive subgoals.

B. Hierarchical Task Planning

Hierarchical approaches reduce long-horizon complexity by organizing tasks around intermediate semantic objectives. HiP [5] coordinates language planning, visual prediction, and low-level control across multiple abstraction levels, whereas DELTA [6] uses LLM-generated subgoals to construct smaller problems for an automated planner. STEP Planner [19] organizes planning through a hierarchical subgoal tree, while HiAgent [7] treats semantic subgoals as working-memory chunks and compresses completed trajectories. Together, these methods establish subgoals as effective units for task decomposition and context organization. However, decomposition alone does not specify how to assess progress from execution evidence for the active subgoal.

C. Feedback-Aware Task Planning

Feedback-aware methods incorporate execution information to improve planning robustness. Inner Monologue [20] integrates environment feedback into closed-loop language planning, while AdaPlanner [9] uses predicted feedback and assertions to refine plans during execution. CoEx [8] verifies subgoal experiences to update an adaptive belief state, and LLaMAR [21] combines subtask verification with execution correction. KnowLoop [10] further detects whether executed subgoals have succeeded using an uncertainty-aware failure detector. Collectively, these methods support plan refinement, belief updating, verification, and failure detection.

In summary, prior work has advanced LLM planning through action generation, hierarchical decomposition, and feedback-based adaptation. However, using explicit subgoal progress to regulate execution remains underexplored. ProLoop addresses this gap through structured state relations.

III. PRELIMINARIES

A. Planning Framework

A task is characterized by a tuple $\mathcal{P} = \langle G, S, O, A, T \rangle$, where G denotes the goal condition, S and O represent the state and observation spaces, A is the set of available actions, and $T : S \times A \rightarrow S$ specifies the state transition induced by action execution. Planning aims to obtain a policy that drives the system from an initial state to a state satisfying G . In LLM-based planning, these task elements are commonly expressed through a natural-language description D provided to the model.

B. Hierarchical Subgoal Execution

In hierarchical LLM planning, a long-horizon task is progressively executed through semantic subgoals. At stage k , the planner generates an active subgoal g_k as an intermediate objective toward the overall goal G . Executing g_k involves multiple environment interactions, forming a local trajectory $\tau_k^t = [(a_{k,1}, o_{k,1}), \dots, (a_{k,t}, o_{k,t})]$. Let m_i denote the information retained from the trajectory of a previous subgoal g_i . The cross-subgoal context before executing g_k is therefore $M_k = [(g_1, m_1), \dots, (g_{k-1}, m_{k-1})]$. Accordingly, the working context of the active subgoal is represented as $W_k^t = [G, M_k, g_k, \tau_k^t]$.

IV. METHODOLOGY

In this section, we detail our proposed framework, ProLoop. As illustrated in Fig. 2, ProLoop augments hierarchical LLM planning with an explicit progress-aware closed loop around each active subgoal. The LLM-Based Subgoal Planner generates an active semantic subgoal and retention cues from the overall goal and confirmed cross-subgoal memory. Structured State Construction represents the desired subgoal outcome and heterogeneous execution evidence under a shared task-state schema. The State-Relation Monitor evaluates each desired condition against the current execution state as *supported*, *unresolved*, or *contradicted*, and aggregates these relations into *ongoing*, *deviated*, or *completed* progress. Progress-Guided Adaptation maps the

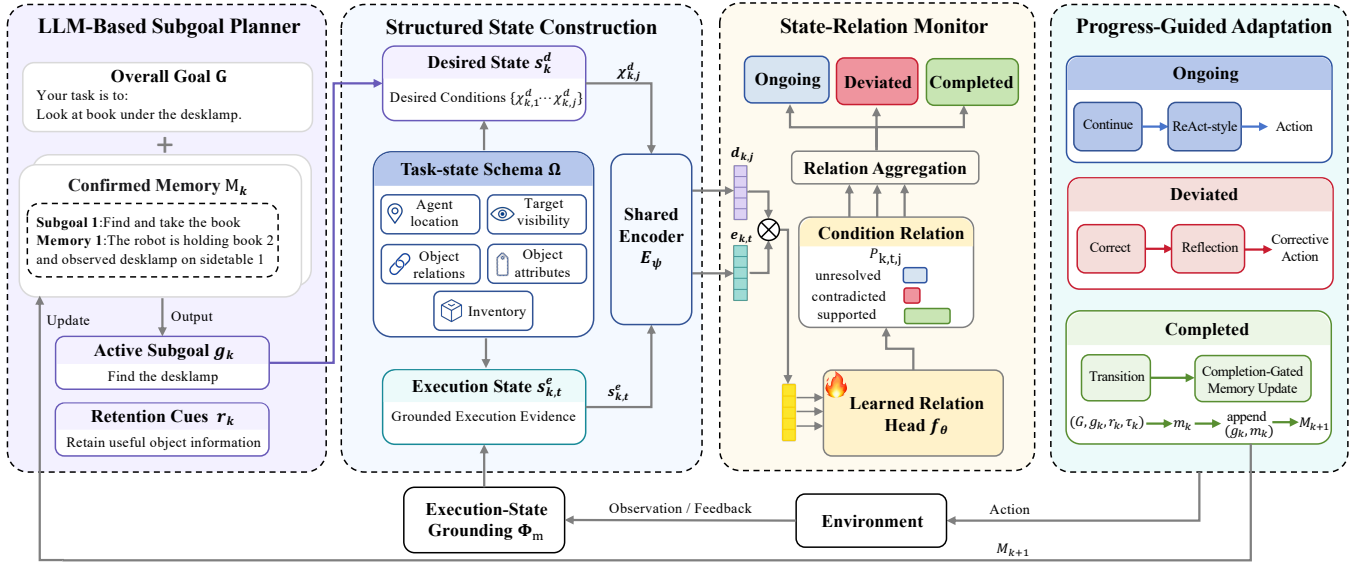


Fig. 2. Overview of ProLoop. Structured State Construction derives comparable desired and execution states for the active subgoal, the State-Relation Monitor derives subgoal progress from their structured relations, and the resulting progress regulates continuation, targeted correction, and subgoal transition.

resulting progress state to continuation, targeted correction, or subgoal transition. Upon completion, Completion-Gated Memory Update uses the retention cues to summarize the local trajectory into a new memory entry before the next subgoal is generated.

A. LLM-Based Subgoal Planner

At stage k , the LLM-based planner generates an active semantic subgoal g_k and retention cues r_k :

$$(g_k, r_k) = P_{\text{LLM}}(G, M_k, I_P), \quad (1)$$

where G and M_k are defined in Sec. III, and I_P contains the planning instruction, in-context demonstrations, and output format. The subgoal g_k specifies the current intermediate objective, whereas r_k indicates information that may be useful to subsequent subgoals. ProLoop generates one active subgoal at a time, allowing later planning decisions to condition on confirmed outcomes from earlier stages.

B. Structured State Construction

Structured State Construction expresses the desired subgoal outcome and the evidence accumulated during execution under a shared task-state schema Ω . It produces a desired state s_k^d and an execution state $s_{k,t}^e$ for subsequent relation assessment by the monitor.

Desired-State Construction. The schema Ω contains general task-state field types, including agent location, target visibility, object attributes, object relations and inventory. Given the active subgoal, an LLM-based constructor generates

$$s_k^d = C_{\text{LLM}}(g_k, \Omega, I_C), \quad (2)$$

where I_C specifies the construction instruction. The resulting state is a sparse set of J_k desired conditions, $s_k^d = \{\chi_{k,j}^d\}_{j=1}^{J_k}$, where J_k is the number of desired conditions for subgoal g_k . Each condition is represented as

$$\chi_{k,j}^d = \langle f_{k,j}, b_{k,j}, v_{k,j}^d \rangle. \quad (3)$$

The three tuple elements denote, respectively, the field type, entity binding, and desired value. Desired conditions may retain category-level bindings, whereas execution states record grounded instances (e.g., mug versus mug 1).

Execution-State Grounding. Let $o_{k,t}$ denote the observation together with available execution feedback returned after action $a_{k,t}$. A modality-specific grounding function extracts newly observed task-relevant fields:

$$\Delta s_{k,t}^e = \Phi_m(a_{k,t}, o_{k,t}, g_k, \Omega). \quad (4)$$

The active subgoal restricts grounding to relevant evidence, while Ω specifies the output schema. The execution state is initialized as $s_{k,0}^e = \emptyset$ and incrementally updated by

$$s_{k,t}^e = \mathcal{U}(s_{k,t-1}^e, \Delta s_{k,t}^e), \quad (5)$$

where $\mathcal{U}$ incorporates newly grounded evidence, replacing updated fields and removing prior facts invalidated by observations or confirmed action outcomes. For constrained text environments, Φ_m uses deterministic event and state parsing. For physical execution, visual observations are grounded by a VLM together with robot and skill feedback. Persistent facts are retained unless invalidated, whereas transient fields are refreshed from current observations, with unavailable values represented as unknown. Let $\text{Inv}(o_{k,t})$ indicate that the execution feedback reports $a_{k,t}$ as invalid. An invalid action is treated as direct contradictory evidence, immediately yielding *deviated* progress and routing the active subgoal to corrective reflection.

C. State-Relation Monitor

As illustrated in Fig. 3, the monitor evaluates each desired condition against the current execution state and aggregates the resulting relations into subgoal progress. Since an unsatisfied desired condition may reflect either unfinished execution or a task-relevant deviation, the monitor distinguishes three relation types. A supported relation indicates

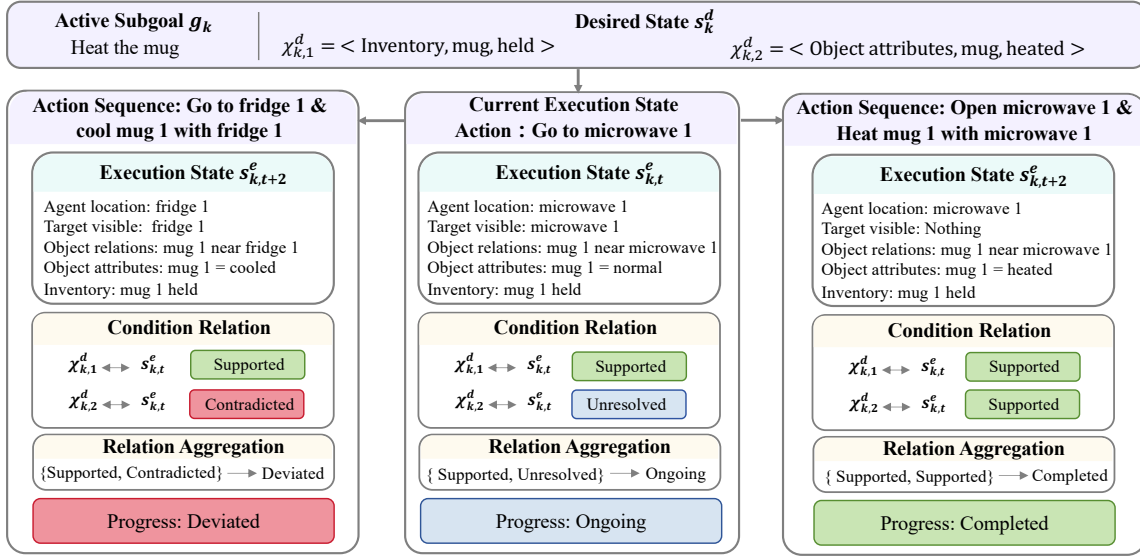


Fig. 3. Worked example of state-relation monitoring and progress derivation. For the active subgoal *Heat the mug*, each desired condition is evaluated against the current execution state, and the resulting condition relations are aggregated to derive *ongoing*, *deviated*, or *completed* subgoal progress.

that the desired condition has been established; an unresolved relation indicates that it is not yet established without a task-relevant conflict; and a contradicted relation indicates a mismatch requiring correction.

The execution state is serialized in a fixed schema order and encoded once per interaction, while each desired condition is encoded individually using the same pretrained semantic encoder:

$$\mathbf{d}_{k,j} = E_\psi(\text{Ser}(\chi_{k,j}^d)), \quad \mathbf{e}_{k,t} = E_\psi(\text{Ser}(s_{k,t}^e)), \quad (6)$$

where $\text{Ser}(\cdot)$ converts a structured input into a canonical textual sequence: execution-state fields follow the fixed schema order, while each desired condition preserves its field type, entity binding, and target value. For each desired condition, we construct a pairwise relation representation using the two embeddings, their absolute difference, and their element-wise interaction:

$$\mathbf{x}_{k,t,j} = [\mathbf{d}_{k,j}; \mathbf{e}_{k,t}; |\mathbf{d}_{k,j} - \mathbf{e}_{k,t}|; \mathbf{d}_{k,j} \odot \mathbf{e}_{k,t}]. \quad (7)$$

Here, $[\cdot; \cdot]$ denotes concatenation and $\odot$ denotes the element-wise product. A shared lightweight MLP relation head f_θ predicts

$$\mathbf{p}_{k,t,j} = \text{softmax}(f_\theta(\mathbf{x}_{k,t,j})), \quad (8)$$

where

$$\mathbf{p}_{k,t,j} = [p_{k,t,j}^{\text{sup}}, p_{k,t,j}^{\text{unr}}, p_{k,t,j}^{\text{con}}]$$

contains confidence scores for *supported*, *unresolved*, and *contradicted*. The predicted condition relation is

$$\ell_{k,t,j} = \arg \max_{c \in \mathcal{Q}} p_{k,t,j}^c, \quad \mathcal{Q} = \{\text{sup}, \text{unr}, \text{con}\}. \quad (9)$$

Subgoal progress is then derived from these relations and

invalid actions:

$$\rho_{k,t} = \begin{cases} \text{deviated}, & \text{Inv}(o_{k,t}) \vee \exists j : \ell_{k,t,j} = \text{con}, \\ \text{completed}, & \neg \text{Inv}(o_{k,t}) \wedge \forall j : \ell_{k,t,j} = \text{sup}, \\ \text{ongoing}, & \text{otherwise.} \end{cases} \quad (10)$$

D. Progress-Guided Adaptation

The derived progress state regulates the underlying executor:

$$\mathcal{A}(\rho_{k,t}) = \begin{cases} \text{CONTINUE}, & \rho_{k,t} = \text{ongoing}, \\ \text{CORRECT}, & \rho_{k,t} = \text{deviated}, \\ \text{TRANSITION}, & \rho_{k,t} = \text{completed}. \end{cases} \quad (11)$$

Continue. For an ongoing subgoal, ProLoop preserves the underlying ReAct-style think-act loop without introducing an additional intervention prompt. The executor continues reasoning from the active subgoal, cross-subgoal memory, and local trajectory.

Correct. For a deviated subgoal, contradicted relations and, when present, invalid-action feedback identify the evidence requiring correction. ProLoop augments the executor's working context W_k^t with this evidence and the current structured execution state $s_{k,t}^e$ through a correction prompt. The executor reflects on the identified mismatch and produces a targeted corrective action without changing the active subgoal. The resulting interaction is evaluated again by the same monitor, yielding updated subgoal progress.

Transition. For a completed subgoal, ProLoop terminates the current execution stage and invokes the Completion-Gated Memory Update in Sec. IV-E to summarize the completed local trajectory into a memory entry. If the overall goal remains unsatisfied, control returns to the planner to generate g_{k+1} ; otherwise, execution terminates. This explicit transition decision avoids relying solely on the LLM's implicit judgment over the accumulated trajectory.

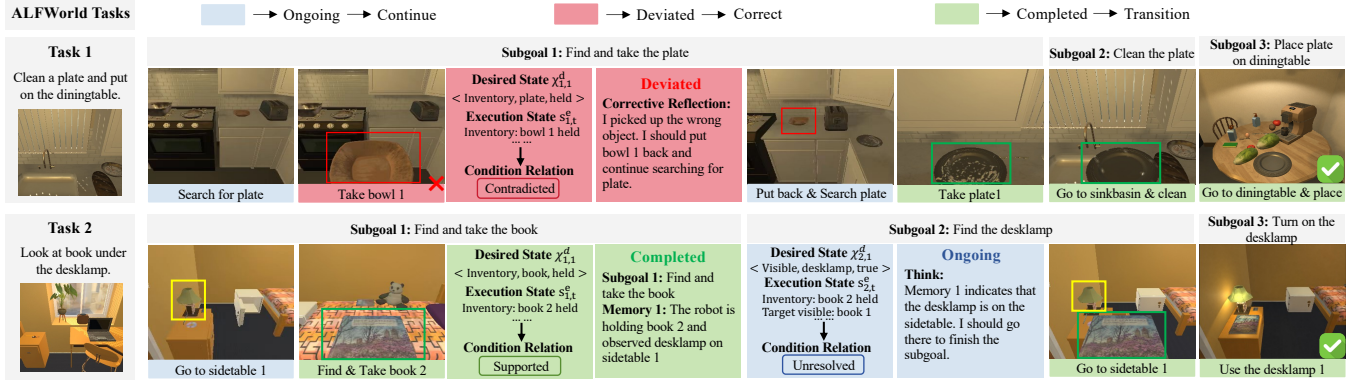


Fig. 4. Representative ALFWorld executions with ProLoop. Ongoing progress allows execution to continue without additional intervention, deviated progress triggers targeted correction, and completed progress enables subgoal transition and memory update.

E. Completion-Gated Memory Update

Cross-subgoal memory is updated only after the derived progress state indicates completion. An LLM summarizes the complete local trajectory τ_k of g_k , guided by the retention cues r_k :

$$m_k = S_{\text{LLM}}(G, g_k, r_k, \tau_k, I_M), \quad (12)$$

where I_M specifies the memory-update instruction. The retention cues guide the summary but do not prevent newly discovered task-relevant information from being retained.

The new subgoal-memory pair is appended to the existing context:

$$M_{k+1} = M_k \oplus (g_k, m_k). \quad (13)$$

Here, $\oplus$ appends a new entry without merging previous memories. Only the cue-guided summary is carried forward; the full local trajectory is not appended to cross-subgoal memory, bounding its size.

F. Offline Relation Learning

The State-Relation Monitor is trained offline using trajectories collected only from training environments. Human judgments define the semantics of the three condition relations through annotated examples and annotation instructions. Guided by this supervision, Qwen3.8-Max serves as the offline annotator and assigns each condition-state pair $(\chi_{k,j}^d, s_{k,t}^e)$ a relation label in $\mathcal{Q} = \{\text{sup}, \text{unr}, \text{con}\}$. A subset of the generated annotations is manually inspected for label consistency. These labels provide condition-level supervision for the relation head; the LLM annotator is used only offline and is not involved during execution.

Given the predicted relation distribution $\mathbf{p}_{k,t,j}$, the relation head is optimized using weighted cross-entropy:

$$\mathcal{L}_{\text{rel}} = - \sum_{k,t,j} \sum_{c \in \mathcal{Q}} w_c y_{k,t,j}^c \log p_{k,t,j}^c, \quad (14)$$

where $y_{k,t,j}^c$ is the annotated target and w_c compensates for class imbalance. We use BAAI/bge-base-en-v1.5 [22] as the pretrained semantic encoder, kept fixed during training; only the lightweight MLP relation head is optimized.

V. EXPERIMENTS

In this section, we evaluate ProLoop on the AgentBoard ALFWorld benchmark under different task difficulties and LLM backbones. The results demonstrate consistent improvements in overall task performance, with particularly pronounced gains on Hard tasks involving more subgoals.

A. Experimental Setup

Environment and Dataset. We conduct our evaluation on the out-of-distribution (OOD) split of ALFWorld provided by AgentBoard [13], [23]. The benchmark contains six household task categories: *Pick & Place*, *Examine in Light*, *Clean & Place*, *Heat & Place*, *Cool & Place*, and *Pick Two & Place*, comprising 134 unseen instances in total. Among them, 24 Easy tasks require three subgoals, while 110 Hard tasks require four. Each episode is capped at 30 environment interactions, and all methods follow the same data split, interaction budget, and evaluation protocol for a consistent comparison across all methods.

Compared Methods. We compare ProLoop with four representative paradigms for LLM-based long-horizon planning: **1) Plan-Only:** BaseAgent, derived from AgentBoard’s VanillaAgent, directly generates environment actions without explicit reasoning or feedback. **2) Reason-Only:** ReAct [15], which interleaves step-wise reasoning and action generation. **3) Reason-Hierarchical:** HiAgent [7], which organizes long interaction histories through semantic subgoals and hierarchical working memory. **4) Reason-Replan:** AdaPlanner [9], which adapts and refines plans using closed-loop execution feedback. We evaluate the methods using Qwen3-14B and GPT-5.6 Luna as LLM backbones. All ALFWorld experiments are conducted over three runs with different random seeds for each method.

Metrics. We evaluate all methods from three complementary perspectives:

- **Overall Task Performance.** Success Rate (SR) measures final task completion, Progress Rate (PR) measures the maximum fraction of benchmark-defined progress steps satisfied during execution, and Grounding Accuracy (GA) measures the proportion of generated

TABLE I

PERFORMANCE OF PROLOOP AND BASELINES ON AGENTBOARD ALFWORLD OOD. RESULTS FOR BOTH BACKBONES ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER THREE RUNS WITH DIFFERENT RANDOM SEEDS.

Method	Overall (%)			Easy (%)		Hard (%)	
	SR $\uparrow$	PR $\uparrow$	GA $\uparrow$	SR $\uparrow$	PR $\uparrow$	SR $\uparrow$	PR $\uparrow$
<i>Model: Qwen3-14B</i>							
Plan-Only	57.71 $\pm$ 2.46	67.77 $\pm$ 1.42	82.02 $\pm$ 1.29	81.94 $\pm$ 1.96	84.26 $\pm$ 1.73	52.42 $\pm$ 3.09	64.17 $\pm$ 1.98
Reason-Only	83.58 $\pm$ 1.06	91.21 $\pm$ 0.44	93.18 $\pm$ 0.38	94.44 $\pm$ 5.20	95.37 $\pm$ 4.72	81.21 $\pm$ 1.87	90.30 $\pm$ 0.95
Reason-Hierarchical	69.90 $\pm$ 1.41	79.71 $\pm$ 1.41	94.53 $\pm$ 0.06	90.28 $\pm$ 1.96	91.20 $\pm$ 0.65	65.45 $\pm$ 1.96	77.20 $\pm$ 1.75
Reason-Replan	75.62 $\pm$ 2.54	83.46 $\pm$ 1.56	91.88 $\pm$ 1.08	100.00$\pm$0.00	100.00$\pm$0.00	70.30 $\pm$ 3.09	79.85 $\pm$ 1.90
Ours	96.52$\pm$0.70	97.39$\pm$0.18	98.23$\pm$0.05	95.83 $\pm$ 0.00	98.61 $\pm$ 0.00	96.67$\pm$0.86	97.12$\pm$0.21
<i>Model: GPT-5.6</i>							
Plan-Only	84.33 $\pm$ 3.39	90.90 $\pm$ 2.08	94.54 $\pm$ 0.25	93.06 $\pm$ 5.20	93.98 $\pm$ 4.29	82.42 $\pm$ 3.09	90.23 $\pm$ 1.70
Reason-Only	87.06 $\pm$ 1.76	92.29 $\pm$ 0.58	96.87 $\pm$ 0.26	93.06 $\pm$ 1.96	93.06 $\pm$ 1.96	85.76 $\pm$ 2.39	92.12 $\pm$ 0.95
Reason-Hierarchical	89.30 $\pm$ 1.53	92.54 $\pm$ 1.19	97.61 $\pm$ 0.33	90.28 $\pm$ 1.96	91.67 $\pm$ 0.00	89.09 $\pm$ 1.48	92.73 $\pm$ 1.45
Reason-Replan	92.54 $\pm$ 1.06	95.52 $\pm$ 0.61	97.34 $\pm$ 0.58	100.00$\pm$0.00	100.00$\pm$0.00	90.91 $\pm$ 1.29	94.55 $\pm$ 0.74
Ours	97.51$\pm$0.35	97.64$\pm$0.18	99.01$\pm$0.28	100.00$\pm$0.00	100.00$\pm$0.00	96.97$\pm$0.43	97.12$\pm$0.21

TABLE II

ENVIRONMENT INTERACTION AND TOKEN EFFICIENCY ON AGENTBOARD ALFWORLD WITH QWEN3-14B. RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION.

Method	Env. steps $\downarrow$	Tokens / episode $\downarrow$
Plan-Only	20.61 $\pm$ 0.47	31,676 $\pm$ 875
Reason-Only	14.98 $\pm$ 0.23	54,713 $\pm$ 2,629
Reason-Hierarchical	17.25 $\pm$ 0.10	35,950 $\pm$ 25
Reason-Replan	17.29 $\pm$ 0.11	8,422$\pm$104
Ours	10.14$\pm$0.13	25,191 $\pm$ 171

actions that are valid under the environment state at the time of execution.

- **Performance by Task Difficulty.** We report SR and PR separately on the Easy and Hard subsets. Following AgentBoard, the subsets are distinguished by the number of required subgoals; Hard tasks contain more subgoals and therefore provide a more demanding evaluation of long-horizon execution.
- **Efficiency.** We report the per-episode averages of environment interactions and total tokens across all online LLM calls during execution.

B. Main Results

Tables I and II report task performance and efficiency, respectively, while Fig. 4 shows representative ProLoop executions. **1) Overall Task Performance.** Compared with the strongest corresponding baselines, ProLoop improves overall SR by 12.94% and 4.97%, and PR by 6.18% and 2.12% under Qwen3-14B and GPT-5.6, respectively. It also achieves the highest GA under both backbones, reaching 98.23% and 99.01%. These results show that explicit progress assessment and adaptation improve reliable long-horizon execution. Notably, ProLoop with Qwen3-14B outperforms all four GPT-5.6 baseline methods in overall SR, PR, and GA, showing that effective progress-aware execution can provide

substantial gains beyond those obtained from a stronger backbone alone. **2) Performance by Task Difficulty.** On the 24 Easy tasks, ProLoop with Qwen3-14B achieves 95.83% SR and 98.61% PR, slightly below Reason-Replan at 100%, while GPT-5.6 ProLoop matches the best result at 100% on both metrics. The advantage becomes more pronounced on the 110 Hard tasks, where ProLoop improves SR by 15.46% and 6.06%, and PR by 6.82% and 2.57% under the two backbones, respectively. Since Hard tasks require more subgoals and transition decisions, the larger gains indicate that explicit progress regulation becomes increasingly beneficial over longer execution horizons. **3) Efficiency.** ProLoop requires only 10.14 environment interactions per episode, the lowest among all methods. Compared with Reason-Only, it reduces environment interactions by 32.31% and token usage by 53.96%. Its 25,191 tokens per episode are the second lowest overall, indicating a favorable balance between task performance and resource efficiency.

VI. ANALYSIS AND DISCUSSION

A. Subgoal Transition Reliability

We conduct a controlled subgoal-transition analysis to examine the effect of explicit progress assessment. The *Implicit LLM* variant corresponds to the *w/o Progress Loop* configuration in Table IV: it retains the ReAct-style executor and cross-subgoal memory, but relies on the LLM to infer subgoal completion from the accumulated trajectory and output COMPLETED for transition. Subgoal transitions from three random seeds are retrospectively labeled using human annotation and LLM-assisted labeling, covering both successful and failed episodes.

As shown in Table III, the Implicit LLM achieves only 68.70% on-time transitions, with 16.08% premature, 11.94% delayed, and 3.29% missed transitions. In contrast, ProLoop increases the on-time rate to 98.66%, while reducing delayed transitions to 1.34% and eliminating premature and missed

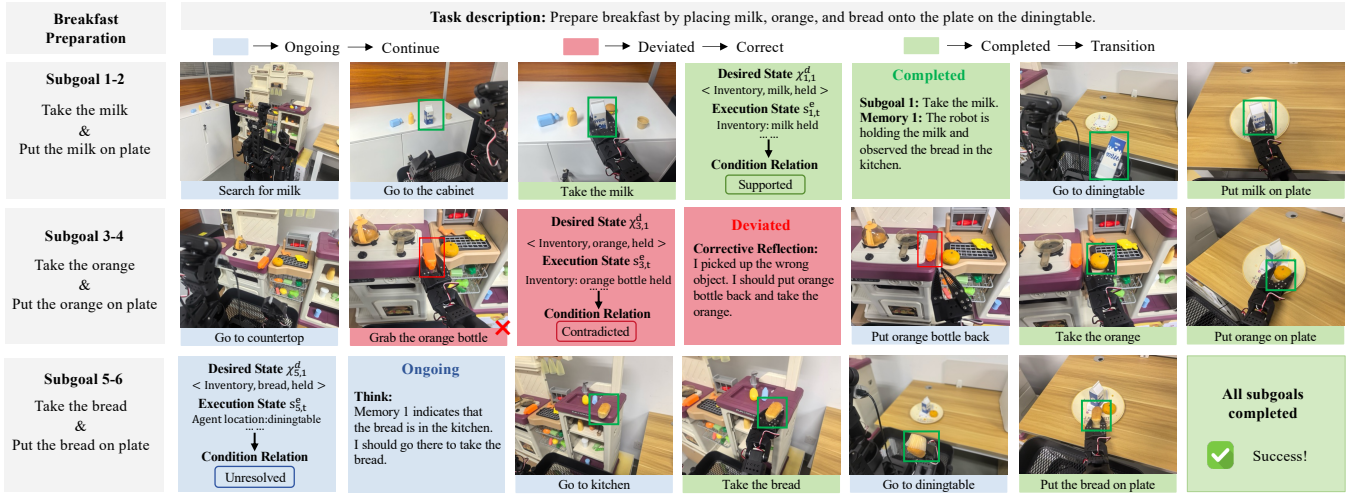


Fig. 5. Successful execution of a breakfast preparation task with ProLoop.

TABLE III

SUBGOAL TRANSITION RELIABILITY UNDER EXPLICIT AND IMPLICIT PROGRESS ASSESSMENT WITH QWEN3-14B.

Method	On-time (%) ↑	Delayed (%) ↓	Premature (%) ↓	Missed (%) ↓
Implicit LLM	68.70	11.94	16.08	3.29
Ours	98.66	1.34	0.00	0.00

TABLE IV

ABLATION STUDY OF PROLOOP COMPONENTS ON AGENTBOARD ALFWORLD OOD WITH QWEN3-14B.

Method	Overall		Easy	Hard
	SR ↑	PR ↑	SR ↑	SR ↑
Ours full	96.52	97.39	95.83	96.67
w/o Progress Loop	85.07	91.92	95.83	82.73
w/o Memory	92.54	96.08	100.00	90.91
w/o Both	83.58	93.97	95.83	80.91

transitions. These results show that implicit trajectory-based completion judgments are prone to inaccurate subgoal boundaries, whereas explicit state-relation assessment enables substantially more reliable transition control during long-horizon execution.

B. Ablation Study

We evaluate the contribution of the two major components of ProLoop in Table IV. The *w/o Progress Loop* variant removes Structured State Construction, State-Relation Monitoring, and Progress-Guided Adaptation. The *w/o Memory* variant disables the Completion-Gated Memory Update, while the *w/o Both* variant removes both mechanisms.

Removing the Progress Loop causes the largest individual SR decrease (11.45% overall), with a large drop on Hard tasks (13.94%) but no decrease on Easy tasks. This suggests

that explicit progress regulation becomes increasingly important as more subgoals and transition decisions must be coordinated. Removing memory reduces overall and Hard-task SR by 3.98% and 5.76%, respectively, while Easy-task SR does not decrease, indicating that cross-subgoal information becomes more valuable over longer execution horizons. Removing both components further lowers overall and Hard-task SR to 83.58% and 80.91%, respectively, supporting the complementary roles of progress regulation and cross-subgoal memory in reliable long-horizon execution.

C. Real-World Robotic Experiment

We further validate ProLoop on a dual-arm mobile robot equipped with a RealSense 435i head camera, using Qwen3.8-Max as the VLM backbone. The robot executes semantic subgoals through four atomic skills: *Grasp*, *Place*, *Move_to*, and *Observe_state*. We evaluate breakfast preparation and tabletop tidying, with 10 trials each. Across 20 trials, ProLoop achieves a 90% overall success rate. Fig. 5 illustrates a representative execution, showing continued execution, targeted correction, and reliable subgoal transition in the physical environment. Additional details and demonstrations are available on our project website and in the supplementary video.

VII. CONCLUSION

We present ProLoop, a progress-aware closed-loop framework for long-horizon robotic task planning. By constructing comparable desired and execution states, ProLoop derives explicit subgoal progress from structured state relations and uses it to regulate continuation, targeted correction, and subgoal transition. Completion-gated memory selectively preserves task-relevant outcomes for subsequent planning. Experiments on ALFWORLD demonstrate improved success rate with fewer environment interactions, while real-world trials further validate effective progress-aware execution on a physical robot.

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